

Artificial Intelligence for RFIC Design: Early Advancements, Opportunities, and Challenges

From discovering nonintuitive design spaces to accelerated RFIC designs

When AlphaGo played its now-famous Move 37 against Lee Sedol in 2016, the moment stood out not simply because it led to a win, but because the move did not resemble the human games the system had been trained on. It was neither copied from precedent nor the result of explicit instruction; instead, it emerged from extensive self-play. The broader implication was subtle but important: The system was not imitating past behavior, but exploring the space of possibilities directly, guided only by the rules of the game. A similar perspective can be applied

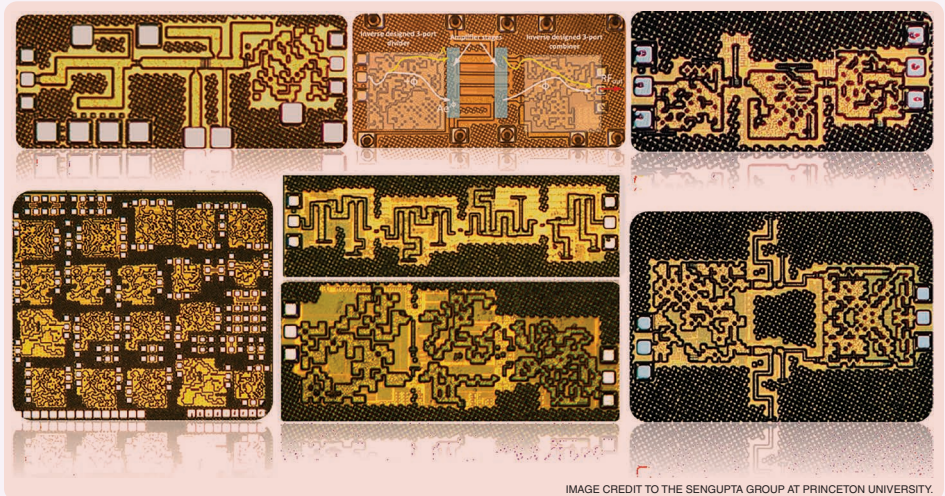


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to RF integrated circuits and analog/mixed-signal design, where the governing physical rules are also well understood, but the design space is highly complex and tightly coupled.

Despite their central role in modern wireless, radar, sensing, and autonomous systems, their design, particularly at RF, millimeter wave (mmWave), and terahertz frequencies, remains

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largely intuition-driven and template-bound. Today's workflows rely on bottom-up assembly, extensive simulation, and iterative trial and error. Unlike digital integrated circuits (ICs), RFICs still lack a scalable, end-to-end automated design flow.

In this article, we present an artificial intelligence (AI)-enabled framework that reframes RFIC design as an algorithmic problem of exploration and synthesis. The approach proceeds along two complementary directions: The universalization of RF passives through inverse electromagnetic (EM) design, and the universalization of RF circuits through a combination of reinforcement learning (RL) and inverse design. Together, these techniques enable rapid, layout- and EM-aware synthesis

of high-frequency circuits directly from specifications, opening regions of the design space that are difficult to reach using conventional workflows. While significant challenges remain, we argue that—if developed thoughtfully—AI has the potential to reshape how RFICs are conceived, explored, and built, shifting the field from template-bound optimization toward more discovery-driven design.

Engineering “Twins”

Engineering progress has rarely been linear. Theory tells us what should be possible, experiments reveal what actually works, and human intuition fills the gap between the two. When intuition reaches its limits, progress depends on new ways

to see the design space: new models, abstractions, or instruments that make previously invisible solutions accessible. Many of the most creative breakthroughs in engineering have not come from better optimization with existing templates of what works, but rethinking the problem from the ground up.

A striking historical example comes from architecture [Figure 1(a)]. For centuries, monumental buildings followed established Gothic and Romanesque forms, refined incrementally through precedent and experience. When Antoni Gaudí began work on the Sagrada Família in Barcelona, he set out to do something fundamentally different: achieve unprecedented height and branching forms without relying on traditional

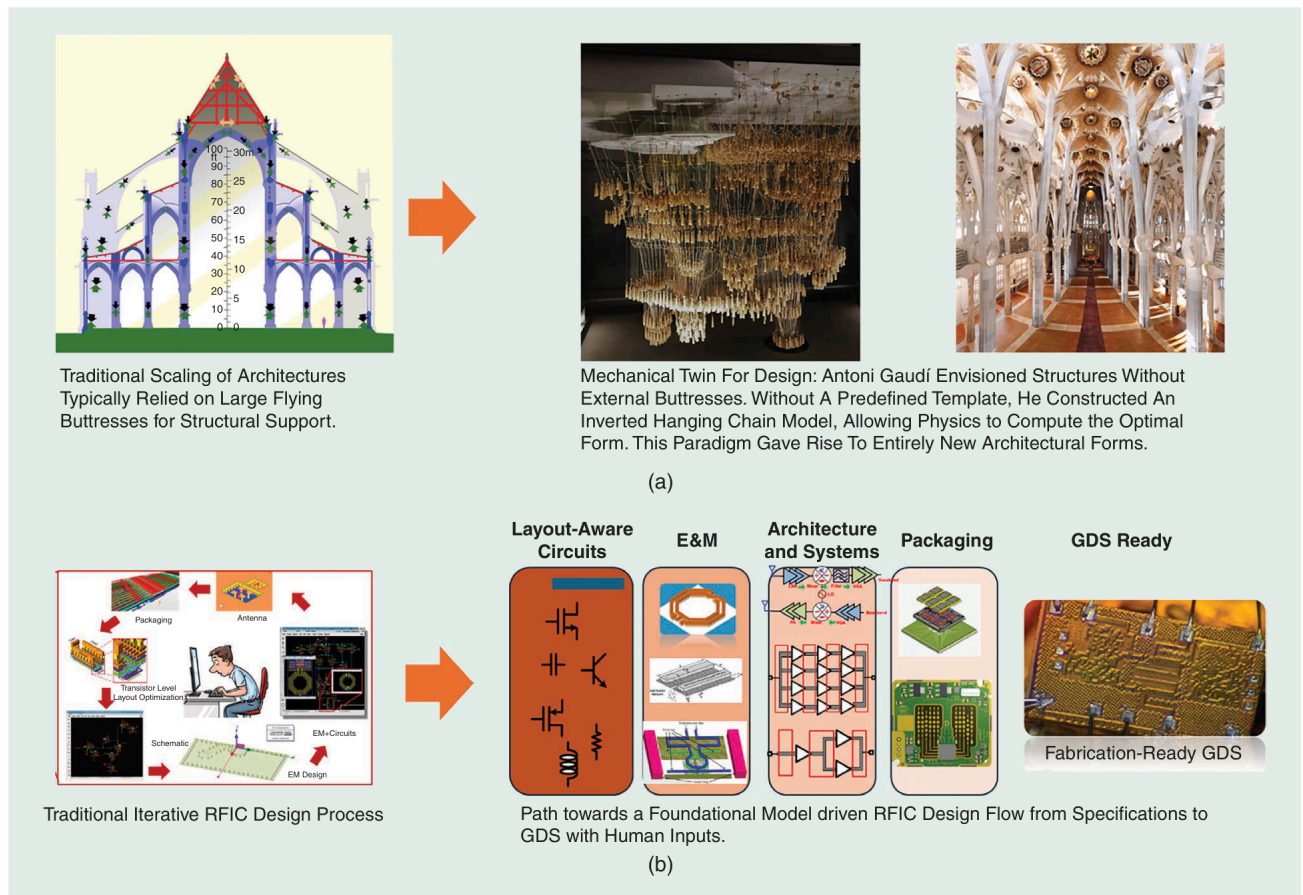


FIGURE 1: Design spaces beyond human intuition (a) Scaling conventional monumental architectures relied on incremental, empirical construction methods, assembling load-bearing elements piece by piece and compensating for unknown force paths through massive walls or flying buttresses. In contrast, Gaudí’s mechanical twin for the Sagrada Familia, using a gravity-driven hanging-chain model, computed the optimal structure using physics of gravity. The mechanical twin opened up a new design space of architectures of tree-like columns and self-supporting vaults, that did not rise from human intuition. (b) A vision in which algorithms shift RFIC design from a traditional, iterative loop, spanning devices, circuits, EM, packaging, and repeated refinement, toward a foundational model-driven approach. In this paradigm, a pre-trained model internalizes the design tradeoff space and makes informed choices across circuit topology, layout, EM structures, architecture, and packaging, potentially through multiple interactions with a human designer, culminating in fabrication-ready chip design.

flying buttresses. It was immediately clear that existing architectural templates would not suffice. But without templates, how does one even begin? How do you identify an architecture template that has never been built before, or define a starting point when the usual rules no longer apply?

Gaudi's breakthrough came from reversing the design process itself. Rather than attempting to calculate the structure analytically or assemble it from known forms, he let physics do the work. He built a gravity-driven hanging-chain model: a mechanical twin that physically computed structural equilibrium. When this model was inverted, the resulting geometry *was* the optimal structure. Out of this process emerged entirely new architectural forms: tree-like columns and self-supporting vaults with no historical precedent.

Without this mechanical twin, such architectures could not have been imagined through traditional bottom-up reasoning alone. More importantly, it revealed a deeper truth: Beyond familiar templates and human intuition lies a vast design space with undiscovered structure and capability, accessible only when we change how we explore it.

"Adjacent Possible": AI in Science and Technology

Advances in AI for RFIC design will not happen in isolation. As Stuart Kauffman described through the idea of the "adjacent possible," certain innovations only emerge when the surrounding conditions are ready. No matter how brilliant the inventor, a motor car could not have appeared in the Middle Ages. The necessary scientific understanding, materials, and infrastructure simply did not exist. The same is true for AI in IC design.

We see a potential inflection point in science and engineering. Protein engineering, of course, has been significantly impacted by AlphaFold and its subsequent iterations, which have more accurately predicted new structures by being able to map

out the underlying interactions in a sequence of amino acids [1]. In fluid dynamics, even though the Navier–Stokes equations have been known for nearly two centuries, turbulence remains an open challenge. Recent, learning-based models have begun to uncover hidden low-dimensional structure, enabling accurate reduced-order descriptions that were previously elusive [2]. In climate science, data-driven approaches, such as GraphCast, can now allow rapid calculations over traditional numerical weather prediction models by learning implicit, multiscale atmospheric dynamics directly from data [3]. In astronomy, machine learning has revealed new classes of celestial objects and exoplanets that classical detection pipelines systematically overlooked [4], [5]. Materials science tells a similar story, where learning-based models have proposed novel compounds (later validated experimentally) that defy long-standing chemical intuition [6], [7]. It is undeniable that these are promising starts, even though it is prudent to remain skeptical of where this progress extrapolates.

RF and analog IC design has the same potential to be positively impacted by advancements in the field. RFIC has long been viewed (correctly) as a blend of physics and intuition. Here, experience guides designers through tightly coupled interactions among devices, circuits, and EM. The approach is typically bottom-up: assembling complex circuits from a small set of canonical circuit and EM templates, refined through ad hoc sweeps and meta-heuristic optimization [8], [9]. While this approach has sustained decades of progress, it implicitly limits the design space inherently to structures that are more easily understood by human intuition [10], [11], [12], [13], [14]. As specifications become more multidimensional and tightly constrained, there is little reason to believe that this assumption still holds.

Moreover, iterative circuit–EM cooptimization is slow, opaque, and rarely generative. It converges on

solutions, but offers limited insight into whether fundamentally different and potentially superior architectures exist. The prevailing belief has been that RF and analog design spaces are simply too complex for systematic algorithmic exploration. Yet this belief mirrors how protein structures, turbulence, weather, and materials were once viewed, until the right representations revealed otherwise [15], [16].

AI in analog and RF design is not new; early efforts date back to the 1970s and 1980s [17], [18], [19], [20]. In tandem with these approaches, automation approaches have ranged from stochastic search [21], [22], [23] to Bayesian optimization, convex programming [24], [25], [26], but these approaches are typically schematic-centric oblivious to layout parasitics. What has changed over the past decade is the emergence of powerful learning frameworks capable of internalizing complex, coupled physics and operating directly on high-dimensional design spaces. In the past decade, there have been some excellent works for machine learning-based RFIC design that uses the machine learning models for rapid optimization [27], [28], [29], [30], [31], [32]. In the past few years, there has been rapid progress in AI for analog design that has leveraged graph neural networks for topology-aware modeling, RL, and large language models. However, true discovery of novel circuits, layouts, and end-to-end workflow (specs to GDS) that produces tape-out ready designs remains difficult [33], [34], [35], [36].

This article asks a simple but consequential question: What can AI-enabled RFIC design truly do—and why should we care? We explore whether modern learning-based frameworks can serve as digital twins for RF and EM design, analogous to Gaudi's mechanical model, not to validate known solutions, but to discover architectures that do not yet exist. Beginning around 2020, we set out to test two core hypotheses: 1) Can AI access RFIC design spaces beyond human intuition? and 2) Can

it reduce design time and effort by orders of magnitude?

Here, we review the progress made over the past five years and outline a deep learning-enabled framework for AI-enabled design of RF-to-THz ICs. We explore how concepts combining inverse design and RL can allow algorithms to navigate this complex multidimensional design space. We would like to explore whether it can generate circuits that take highly nonintuitive forms, accessing regions of the design space that are difficult to reach with classical workflows, and much more rapidly than traditional design methods. This framework builds on our initial demonstrations in 2022 [37], [38], and the rapidly growing body of related work [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54]. Like autonomous driving, AI in circuit design spans an L1–L5 spectrum, with L5 representing true spec to graphic-design-system (GDS) autonomy. We are still early on this path, but the real impact is not just faster design. The element of discovery through algorithms should be a major focus.

RFIC: Circuit and EM Codesign—Juggling Decision Across Architecture, Circuits, and EM to Design Within Constraints

At its core, RFIC performance is governed by a deeply nonconvex design space shaped by two inseparable axes: 1) architecture including circuit topology and EM structures, and 2) the continuous parameters that describe them. The codesign between EM and circuits is what fundamentally distinguishes RFIC design from other circuits. As illustrated conceptually in Figures 2 and 3, a significant portion of RFIC design starts from a familiar set of circuit and EM structures and then combines them to form different circuits, and then further tune parameters within those choices.

Consider the seemingly simple task of designing a 60-GHz mmWave power amplifier (PA) in a SiGe BiCMOS (or advanced CMOS) process,

targeting 20-dBm output power at roughly 25% peak efficiency. Even with this minimal specification set, the design does not reduce to a clean optimization problem. After rough hand calculations suggest a plausible device size, the real decisions only begin after taking into consideration several tradeoffs and constraints that are often conflicting.

Should the output power be generated by a single large device, or distributed across multiple smaller devices operating in parallel (Figure 2)? If the latter, how many paths should be combined, and at what point do combining losses erase the efficiency gains from favorable parasitic scaling? For lower

output power levels, such splitting may be unnecessary or even detrimental. Where, exactly, is the crossover point? What output stage configuration is appropriate given the supply voltage and frequency: common-emitter, common-base, cascode, stacked devices, or some hybrid thereof? Each option has different advantages at certain frequencies and bias conditions, and each introduces different EM sensitivities.

The complexity compounds quickly. If cascodes or stacks are used, how many devices should be stacked? How many gain stages are required to meet the overall gain target, and should a single driver feed multiple output cells, or should each cell be

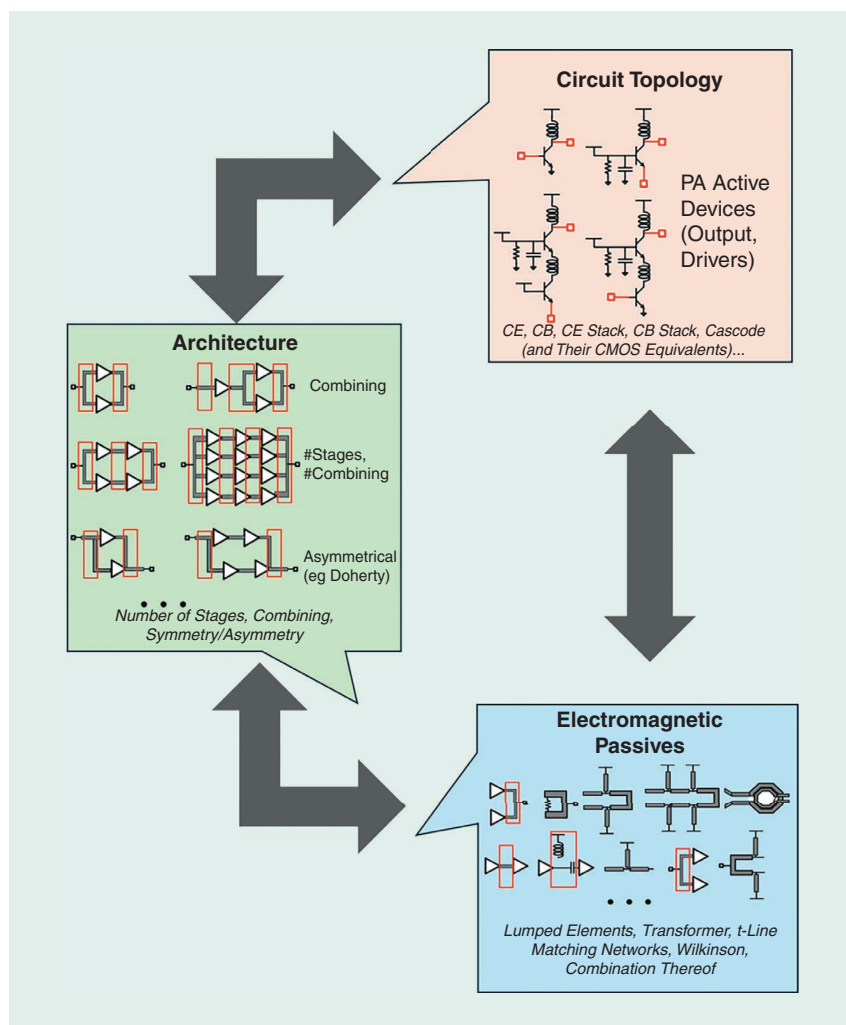


FIGURE 2: RFIC design is an inherently coupled codesign problem spanning architecture, circuit topology, and EM structures. The combinatorial space formed by these choices is far too large to explore exhaustively, and closed-form analytical descriptions are rarely available. As a result, designers typically build intuition through iterative trial and error, navigating a vast but still template-limited subset of known circuits and EM structures.

driven independently (Figure 2)? None of these choices can be made in isolation. Each commits the design to a distinct architectural path, tightly coupled to layout parasitics, matching-network realizability, and EM losses.

The space of possible passive realizations further amplifies the challenge. Output networks can be implemented using lumped elements, transformers, transmission lines, Wilkinson of multisection combiners, or innumerable combinations thereof. There is no closed-form answer to what is optimal. The only way to know is to design the circuit, lay it out, synthesize the EM structures, and cosimulate, often repeatedly.

Only after this iterative process does the designer get a clearer picture on which design decisions to choose, guided by the specification and constraints one has to work with. Optimal device widths, bias points, matching networks, and EM geometries emerge. Faced with this combinatorial explosion, most designs converge not on what is globally optimal (if there is), but on what is familiar, reliable, and known to work in a given process. Through this trial-and-error process, designers gradually build intuition, but that intuition is technology-specific. When the process design kit (PDK) changes, much of that knowledge must be relearned.

Surrogate or forward models can accelerate parts of this workflow when

the topology is fixed [18],[28], but they do not change the fundamental limitation: Architectural choices are still made upfront, and significant redesign is often required once packaging and integration effects are introduced. True specification-to-GDS automation remains elusive because it demands models that can reason across abstraction layers, simultaneously spanning architecture, topology, circuit parameters, EM geometry, and their mutual coupling.

Stepping back, an even more fundamental question emerges. Why are we confined to this space of known circuits and architectures at all? Today's RFIC design space is largely bounded by variations and combinations of a small set of established topologies and EM structures, along with their well-understood limitations. Is there a way to move beyond this inherited design space? That question sits at the heart of AI-enabled RFIC design and motivates the search for methods that do not merely tune templates, but expand the space of architectures that can be explored.

EM Structures Are Simply Ways to Store and Guide Electric and Magnetic Energy—So Why Should They Be Confined to a Handful of Familiar Shapes?

The obvious answer is that human minds think inductively. We understand how lumped elements, such as L and C behave; from there, distrib-

uted transmission lines, essentially cascades of L-C sections, are easy to understand. Once we understand a single transmission line, we naturally understand what can be done with two, three, or four lines, leading to matching networks, combiners, splitters, or hybrids [8]. Over the years, pioneers in RF and microwave design have invented various templates for each of these functions, and these have become the standard library from which designers choose (Figure 3).

But is it possible that beyond these regular structures there exists a host of more complex EM structures that we simply do not know how to design: structures that do not resemble a Lego-like assembly of transmission lines, yet are potentially better? By “better,” we might mean more compact, higher efficiency, broader bandwidth, or improved performance with respect to whatever design objective is most critical.

More fundamentally, the templates we rely on are limited in what they can express, particularly when it comes to narrowband assumptions. For example, quadrature hybrids with four t-lines are known. But, what if we need a dual-band or triple-band hybrid? What if we need a dual-band hybrid whose phase relationship is not fixed at 90°, but varies with frequency to compensate for the parasitic effects of the transistor driving it? What if we need a dual-band PA for a multiband Doherty architecture? How do we synthesize such structures? Typically, these problems are approached using heuristic or intuition-driven design strategies. In contrast, a seemingly arbitrary EM structure can achieve the same functionality, often more compactly and efficiently, without being constrained by predefined templates.

Even Templates Are Not Easy to Design

Figure 4 illustrates representative matching-network examples for PAs designed in a 90-nm SiGe BiCMOS process, highlighting how passive design choices directly influence

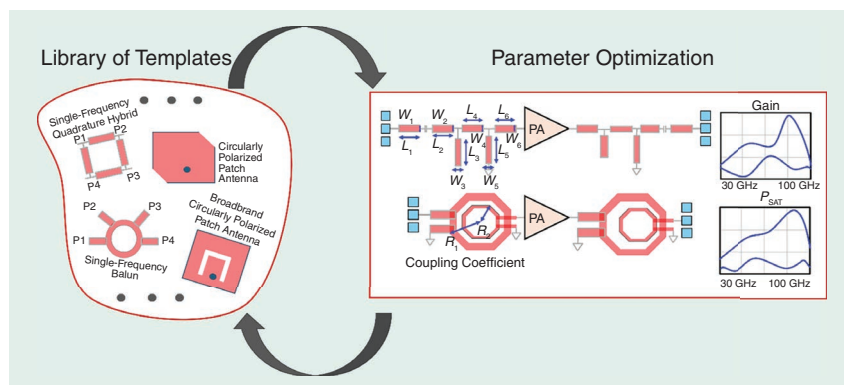


FIGURE 3: The starting point of a RFIC design is often choosing a circuit and EM template, and then parameter optimization is carried out to investigate if specifications can be met. If not, the template library is revisited and the process is repeated, along with designer's own innovation if necessary until specifications are met.

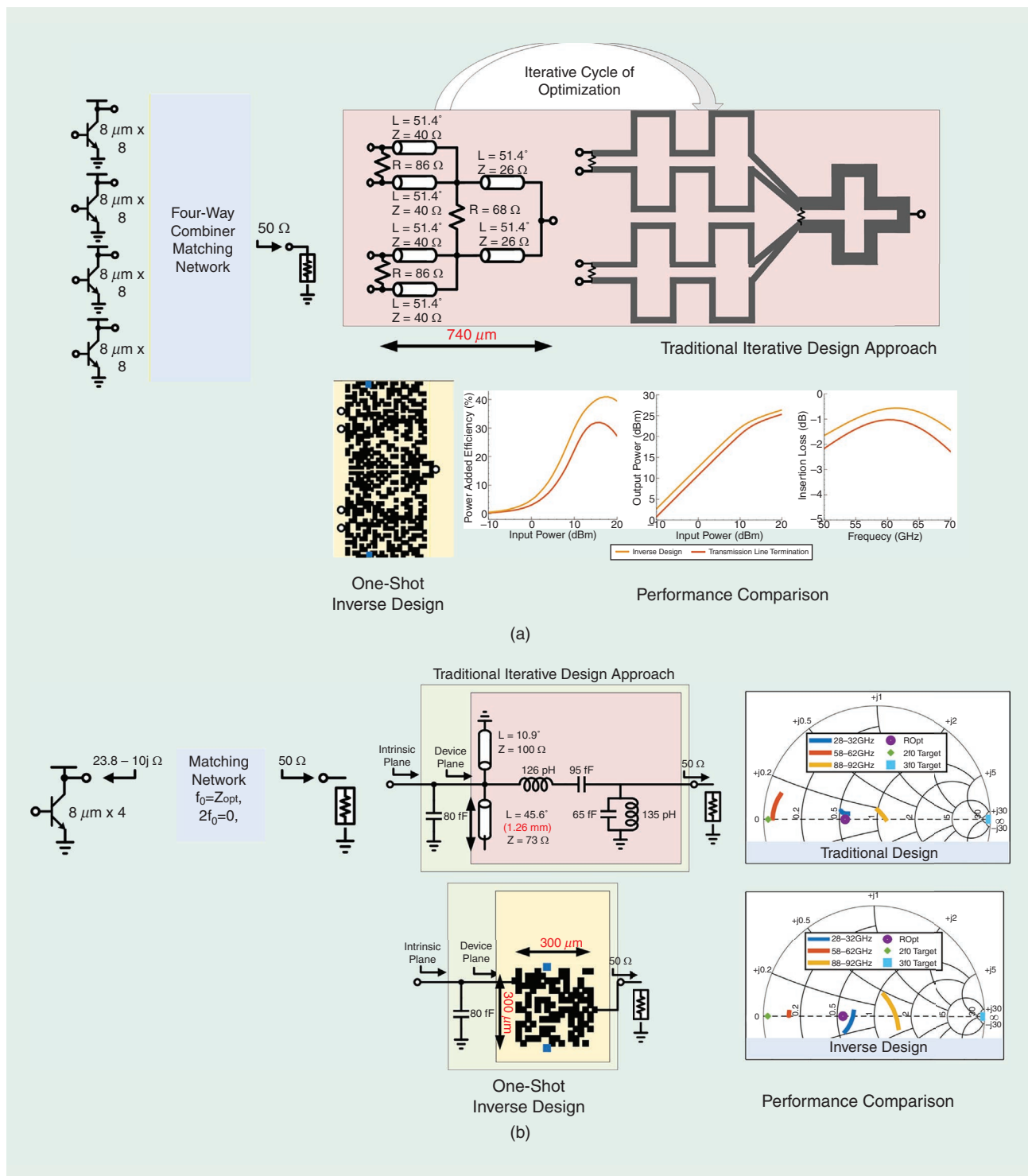


FIGURE 4: RF circuits employing inverse-designed passives and their implications for template-based (a) and template-free (b) design. (a) A four-way combined 60 GHz PA output stage implemented in a 90-nm SiGe BiCMOS process using fully laid-out and extracted transistor models. The design and physical realization of a classical Wilkinson combiner to achieve the required EM performance within a constrained area is nontrivial and typically requires iterative design and optimization. In contrast, inverse design enables a one-shot synthesis that produces a more compact and efficient structure. (b) The greater advantage emerges in template-free RF circuits. Consider a 30 GHz PA incorporating a second-harmonic short, where constructing loss-aware lumped or transmission-line templates is particularly challenging, and achieving the required multi-frequency impedance profile is not straightforward. The inverse-designed structure, synthesized in minutes, occupies a compact footprint and directly realizes the desired response. While the conventional topology shown is not necessarily globally optimal in terms of loss and bandwidth, identifying such optima through exhaustive search is impractical.

circuit performance and architectural decisions. In Figure 4(a), a four-way power combiner achieves nearly 25 dBm of output power at 60 GHz. All simulations are performed using extracted transistor layouts and full EM analysis, ensuring that both parasitic and EM effects are fully captured. While a conventional multistage Wilkinson combiner can be used for narrowband operation, translating its schematic into a practical layout is often nontrivial. The process requires careful routing, line bending, and junction placement to meet area and connectivity constraints, typically involving multiple rounds of iteration. In contrast, the arbitrary-shaped EM structure shown here achieves higher efficiency with significantly less design effort. This example highlights a case where templates exist, yet practical implementation remains challenging.

In many situations, suitable templates do not exist at all. Consider a 30-GHz PA requiring a shorted second-harmonic termination at 60 GHz. Once loss, coupling, layout area, and multifrequency constraints are considered, it is unclear what the optimal solution should be. While prior work has derived globally optimal impedance terminations for single-frequency cases with lossy lumped elements [16], no closed-form solutions exist for the more general,

layout-aware, multifrequency problem. Figure 4(b) compares a traditional lumped and transmission-line-based implementation with an inverse-designed EM structure. Although the template-based network achieves the desired impedance transformation, it incurs nearly 3 dB of loss. The inverse-designed structure, by contrast, occupies a compact footprint and exhibits only 0.5 dB loss, while simultaneously achieving the optimal load-pull condition at 30 GHz and a short at 60 GHz.

More importantly, there exists a much broader class of circuits for which reliable templates are far more difficult to identify. Consider, for example, the design of a dual-band or triple-/multiband power amplifier with a prescribed frequency response that also provides harmonic terminations at the appropriate frequencies to improve efficiency, or a truly broadband PA. In such cases, there is no obvious starting point or canonical structure. Readers are encouraged to read [56] and [57] for examples of highly complex, multistage transmission-line-based networks developed for dual-band PAs. Inverse design can open up new design possibilities in such cases.

Now imagine a design flow in which the input is not a hand-crafted topology, but a set of desired scattering parameters. The output is an EM structure synthesized automatically, while

still respecting designer intent and practical constraints. This shift, from choosing structures to inferring them, is what we refer to as *inverse design*.

AI-Enabled Inverse Design Discovery and Thinking Beyond Templates

Inverse design is the first step toward what we call *universalization*. As illustrated in Figure 5, rather than committing to predefined EM templates, we enable algorithms to synthesize EM layouts directly from target S-parameters for any number of ports [35], [36], [37]. The resulting geometries are often irregular and counterintuitive, but they can outperform hand-designed structures for their intended roles. A single framework can now produce a two-port matching network, a splitter, a hybrid, or a multiport coupler, without ever being told what those components are “supposed” to look like. By removing templates, the design space becomes constrained only by Maxwell’s laws, opening possibilities that conventional RF practice cannot even express. They do not violate the fundamental limits [16], but they open a new design space beyond templates.

Once passive synthesis is freed in this way, a natural question follows: Can the same idea extend to complete RFICs? Can active devices and arbitrarily shaped passives coevolve to form entire circuits? As shown in Figure 5, this generalized formulation allows PAs, oscillators, low-noise amplifiers (LNAs), mixers—and eventually full transceivers—to emerge from the interaction between device physics and EM structure, rather than from predefined architectures [37], [38], [49]. The resulting circuits may look unfamiliar, but they span a far richer space of RF possibilities. Inverse design in spirit of this principle was previously pursued purely through intuition-driven solutions in [58], [59], [60], and is therefore limited in its scope.

Of course, this freedom comes at a cost: scale. Even a modest 25×25 -pixel EM layout corresponds to an astronomically large design space (roughly 10^{180} possible configurations), far

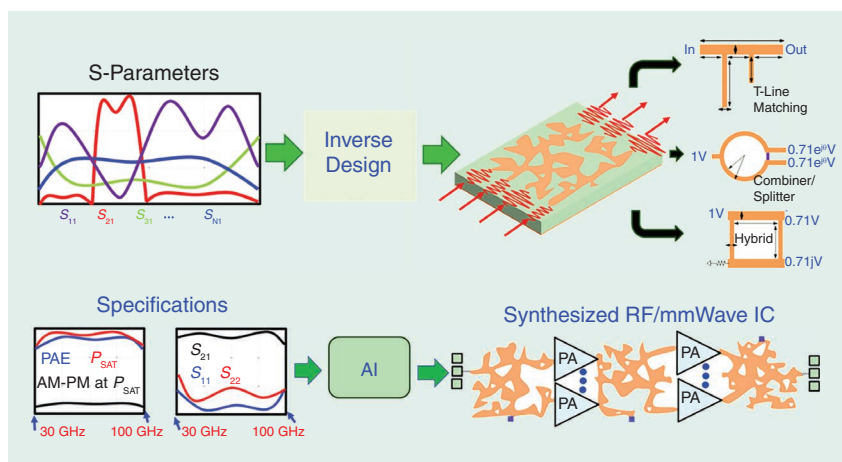


FIGURE 5: Universalization of RFICs where actives and passives get connected together, template-free, to create end-to-end optimal performance. This approach, thereby, allows synthesis of geometries and architectures for which there are no templates, which is a much larger space compared to where templates exist [38].

beyond brute-force search. Classical metaheuristics struggle here, not only because they are slow (minutes to hours per evaluation), but because they do not learn. Every new design starts from scratch, with no accumulated understanding of the design space. Fast and efficient numerical solvers and optimization are certainly an important direction worth pursuing, [63], [64]. However, in our view, the real opportunity extends beyond numerical speed alone. A model that can internalize prior experience and use it to inform subsequent decisions—much like a human designer—deserves serious consideration.

To overcome this barrier, we replace simulation-centric exploration with learning-centric design. At the core of the framework is a deep-learning-based forward model that predicts the EM response of arbitrary on-chip 2D structures. As shown in Figure 6, an AI-based forward model (convolutional neural network or other networks) is trained to map geometry directly to S-parameters, capturing both local features and long-range EM interactions [37], [38]. Once trained, the model evaluates new layouts in milliseconds, compressing what would normally require hours of EM simulation into near-instantaneous inference. This capability fundamentally reshapes the workflow, enabling rapid and broad exploration of a previously inaccessible design space.

Inverse Design Is Inherently Non-unique: A One-to-Many Mapping

Accurate prediction of absolute behavior capturing both mismatch and loss is essential for synthesis. Figure 6 shows error distributions across the 30- to 100-GHz band, with predictions tightly centered around zero, indicating that the surrogate model can reliably replace full-wave solvers. Efficient training is achieved using rapid simulations in air, followed by transfer learning to dielectric substrates [38].

With a fast and accurate forward model in place, the design paradigm itself changes. Instead of proposing

Once trained, the model evaluates new layouts in milliseconds, compressing what would normally require hours of EM simulation into near-instantaneous inference.

geometries and observing their responses, we can now start from desired S-parameters and synthesize the structures that realize them. This inverse-design concept is illustrated in Figure 7. While a given structure has a unique response, many different geometries can realize the same behavior [64], making inverse design inherently nonunique. Several approaches address this challenge. Early work embedded the surrogate model within fast optimization loops [37], [38], [49], replacing costly EM simulations with millisecond-level inference. Subsequent efforts used inverse neural networks to map target S-parameters directly to geometry [52], [55]. More recently, we demonstrated generative approaches based on controlled diffusion [50], similar to how images are generated from text in generative AI.

Figure 8 illustrates the training time reported in [37] and [38] inverse design time compared to a running brute-force genetic algorithm and

iterative EM simulations. What could have taken days can be reduced to minutes of synthesis time, after a one-time investment in training.

Inverse Designed RFICs

The resulting structures are shown in Figure 9. Compact two- and three-port mmWave filters, broadband power splitters, and duplexers emerge with performance difficult to achieve using classical templates. By conditioning the generative process, designers can steer layouts toward transmission-line-like forms, maze-like geometries, or fully free-form structures, depending on the application.

Crucially, these inverse-designed passives integrate naturally into RFICs. In early demonstrations [37], [38] we used inverse synthesis to design broadband matching networks for mmWave PAs spanning 30–100 GHz (Figure 9). While the active devices were designed conventionally, their broadband load-pull targets were provided to the inverse-design engine. The resulting design

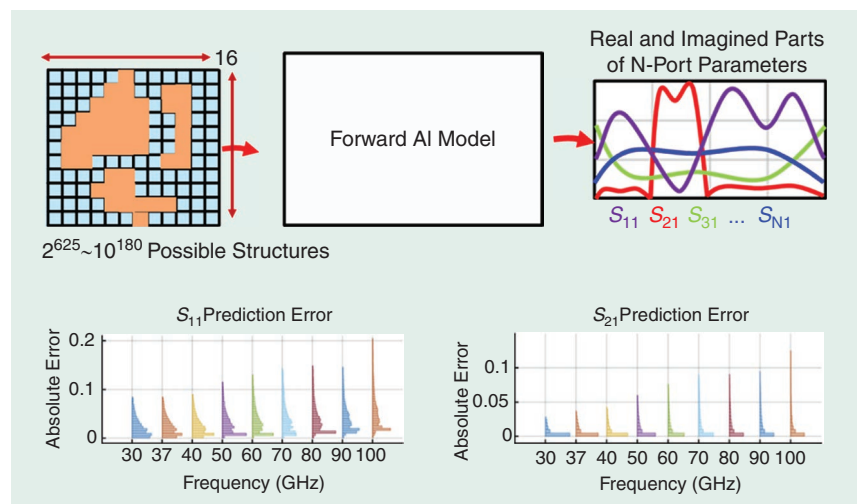


FIGURE 6: Forward pretrained model for rapid inferencing of scattering parameters for any shaped EM structure. The rapid millisecond level inference times allows navigation of a vast design space, by eliminating the need for time- and resource-intensive EM simulations [38].

rule check-compliant but highly unconventional matching networks enabled record bandwidth and efficiency for silicon mmWave PAs at the time (Figure 9).

This approach extends to full multiport architectures. In later work [49], inverse-designed dividers and combiners were used to synthesize complete multipath amplification chains (Figure 10). During end-to-end optimization, the algorithm independently learned that equal phase shifts in the two paths were required for maximum efficiency without being explicitly instructed and despite asymmetric final layouts [49]. Because passives

often constrain RF and mmWave architectures, the ability to synthesize EM structures directly from specifications enables circuit topologies that are difficult, or impossible, to realize using conventional design flows.

The concept of inverse design has since been broadly demonstrated by multiple research groups, showing improved performance and higher levels of automation for RFIC design [40], [41], [42], [43], [44], [45], [46]. For example, as shown in Figure 11, [43] employed inverse design to synthesize a multilayer structure enabling a high-efficiency 6G FR3 PA. In [41], a

full nonintuitive RFIC was optimized directly using meta-heuristic algorithms and iterative EM simulation, achieving the target PA performance through fully automated synthesis (Figure 11).

Inverse Design Through Generative AI With Designer Inputs

It is important to emphasize that the synthesized structures do not need to be arbitrarily shaped. As discussed earlier (see “Inverse Design Is Inherently Nonunique: A One-to-Many Mapping”) the inverse-design problem is inherently ill-posed: Many different structures and even different circuits can realize essentially the same electrical performance. This nonuniqueness can be turned into an advantage. Rather than allowing the algorithm to generate unconstrained geometries, it can be guided to produce structures that look more familiar or adhere to designer preferences.

In [50], we introduce the concept of controlled diffusion, inspired by generative diffusion models used in image synthesis. Within this framework, the generative process can be steered toward different structural priors. As illustrated in Figure 12, passives with identical performance can be synthesized in multiple forms, ranging from maze-like transmission-line structures to more pixelated layouts. Importantly, these same ideas extend beyond individual passives. As shown later, controlled diffusion enables the synthesis of complete RFICs with designer-controlled EM structures, combining algorithmic discovery with human-guided architectural intent. Together, these results present promising examples to explore beyond predefined templates to a new design space, enabled by AI-enabled end-to-end RFIC design: How to choose architecture, circuit topologies and EM all together?

Let us return to the question behind Figure 2: How does an RFIC designer choose the topology, the passives, and the overall architecture? In the real world, those decisions are

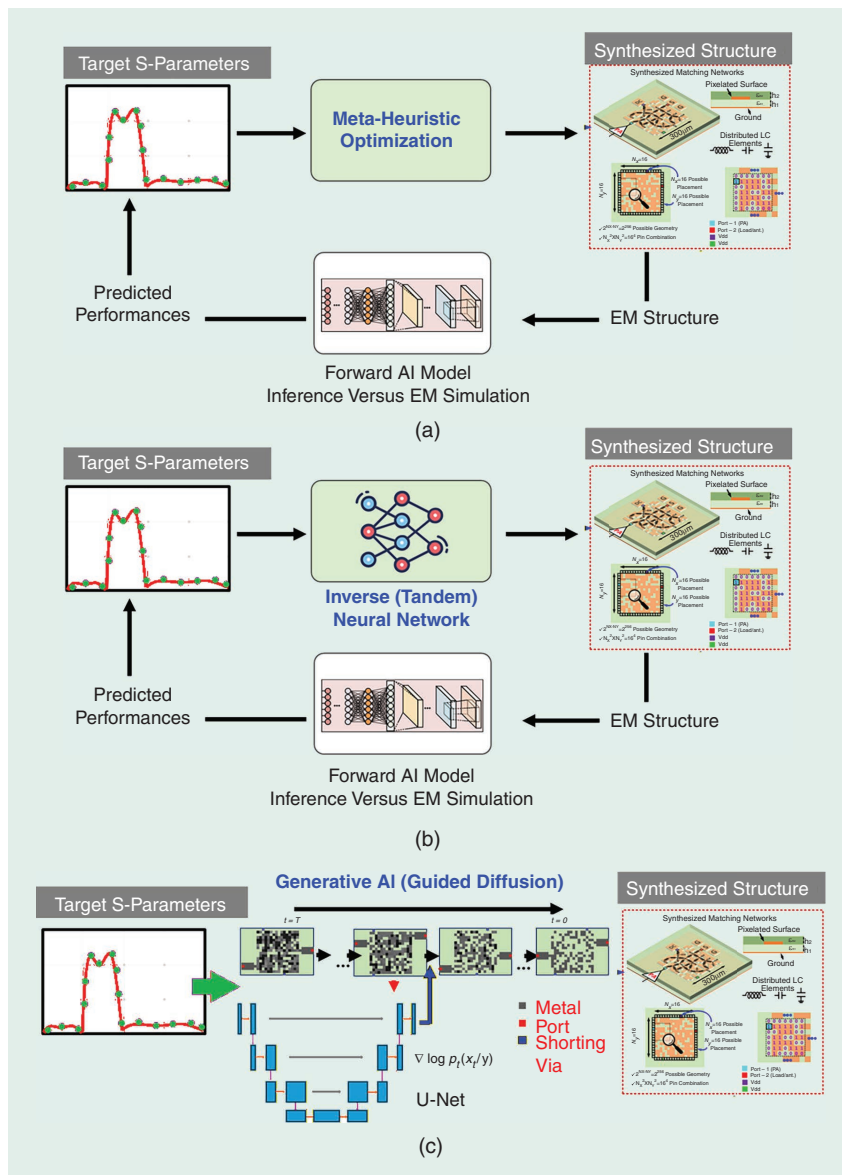


FIGURE 7: Inverse design with the AI-based forward model and utilizing either metaheuristic optimization (a) [28],[37], (b) tandem neural networks [52], or (c) generative diffusion models [50].

not derived from a single equation. They come from an internalized mental model of what on-chip EM can and cannot do, such as enumerating how much bandwidth a matching network can realistically support, how loss scales with geometry, and where parasitics will dominate. Expert designers build this intuition the hard way: years of design with EM behavior, layout-dependent effects, and circuit-level behavior.

Could a model be trained to internalize the same design space

(including its constraints, opportunities, and tradeoffs) and use it to make architectural decisions systematically? If so, it could do more than speed up a workflow. It could provide principled guidance to less experienced engineers, and perhaps even generate architectures that are rarely explored because they fall outside the standard templates. The endgame is ambitious: an algorithm that infers the full RFIC architecture from a specification, such as selecting topologies, sizing devices, setting

bias conditions, and synthesizing the required EM interfaces. It is also a formidable challenge.

As illustrated in Figure 13, one practical route toward this goal is to treat RFIC design as a sequential decision problem and cast it in a reinforcement learning (RL) framework [51]. In this formulation, the “state” spans multiple abstraction levels at once: architectural choices (e.g., number of stages, number of combining paths), circuit configurations (e.g., common-source/emitter, common-gate/

	Dataset	Time Per Simulation	Number of CPUs	Memory Utilization Per Simulation	CPU Efficiency	Number of Simulations	Total Computation Time
Transfer Learning	$(300 \times \epsilon_{\text{eff}}) \times (300 \times \epsilon_{\text{eff}}) \mu\text{m}$ ($\epsilon_r=1$)	~80 Seconds	1	1 GB	100%	250K	Roughly 14 Hours on Princeton HPC (400 CPU Core)
	$300 \times 300 \mu\text{m}^2$ (ϵ_{eff} for chip dielectric)	~907 Seconds	2	3 GB	83%	75K	Roughly 4 Days on Princeton HPC (400 CPU Core)
	$200 \times 200 \mu\text{m}^2$ (ϵ_{eff} for chip dielectric)	~431 Seconds	2	2.4 GB	83%	15K	Roughly 18 Hours on Princeton HPC (400 CPU Core)
Synthesis of EM Structures	Inverse Synthesis (~4,000/40,000/400,000 Population Size and 100 Generations i.e. 400K/4M/40M Total Structures)	~0.3 Seconds for a Batch of 4096	32 (+1 CPU)	<32 MB	>75%	400K/4M/40M (Predicted)	400K~400 Seconds 4M~1 hour 40M~11 hours (Generalizability: Same/Similar Training Set Applicable for any EM Structure, Such as Filters, Matching Networks, Couplers, Hybrids, Antennas etc.)
	Genetic Optimization with Electromagnetic Simulations (~4,000/40,000/400,000 Population Size and 100 Generations i.e. 400K/4M/40M Total Structures)	~907 Seconds	2	2.4 GB	83%	400K/4M/40M	410K~21 days 4M~210 days 40M~2100 days (Generalizability: Need New Optimization for Each Structure)

FIGURE 8: The figures shows that for a given population size for genetic algorithm (in this case), days/weeks of optimization can be reduced to minutes [38].

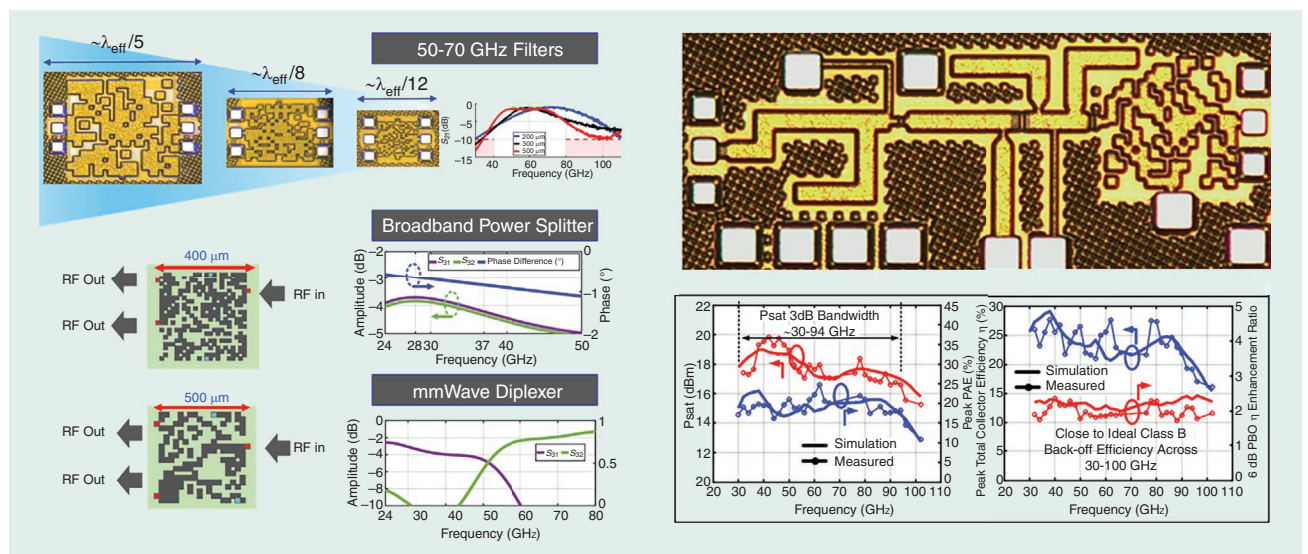


FIGURE 9: Inverse design of two-port, three-port EM structures and broadband 30- to 94-GHz PA with inverse designed output matching network [37], [38], [49].

base, cascode, stacked variants), desired interface impedances (from load-pull, noise matching, or power

transfer objectives), and continuous parameters, such as device geometry and biasing. The resulting search

landscape is highly nonconvex and tightly coupled, exactly the kind of space where local tuning often fails to find good global solutions. We train an RL agent to navigate this EM- and layout-aware space, as illustrated in Figure 13.

Two design principles are central. First, the agent is trying to learn the design space, where architecture and topology choices converge under a multiobjective reward that encodes the target specifications. Second, training is performed with laid-out devices and EM-aware models, so that layout parasitics and passive losses are part of the learning loop. In other words, when the agent converges, the solution does not require the typical postlayout retuning cycle.

Operationally, the agent proposes iterative modifications, such as changing architectural decisions, selecting configurations, adjusting device sizes and bias points. It receives a reward based on how well the evolving design

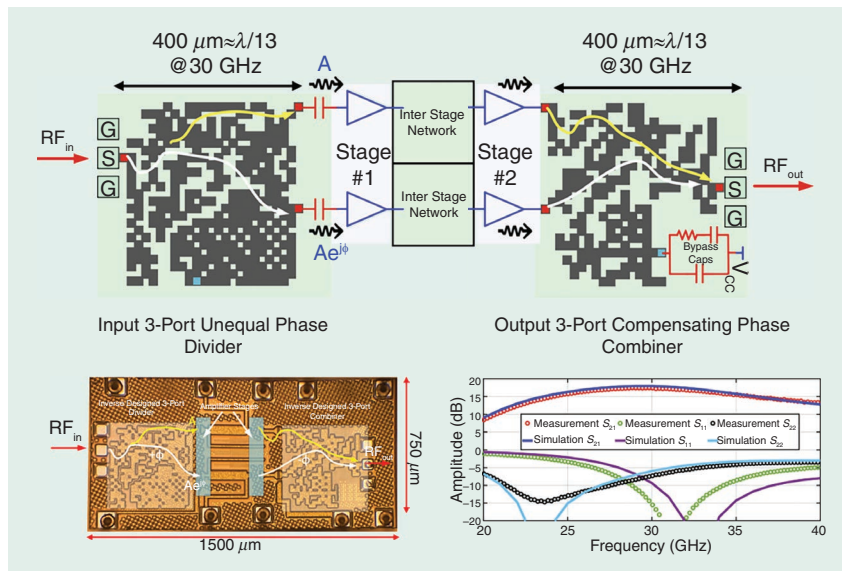


FIGURE 10: AI-driven synthesis of multiport networks and amplifiers reveals an autonomously discovered, nonintuitive topology that enables asymmetric signal splitting and combining [49], aligning with prior observations that asymmetry can yield superior performance in wideband RF systems [10], [60].

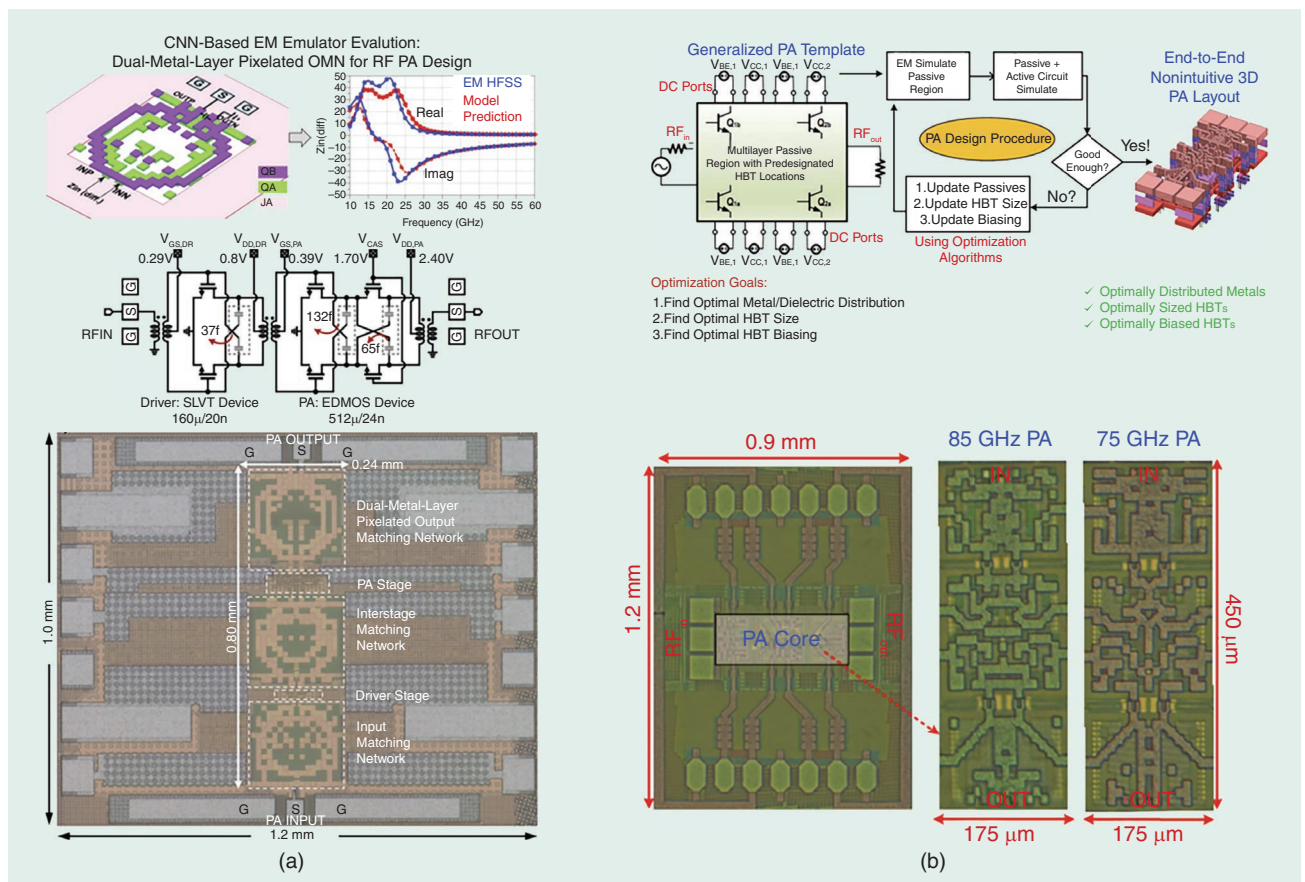


FIGURE 11: Inverse design enabled PAs demonstrated with multistack metal layers (a) [43] and (b) [41].

meets the specification. Once an architecture stabilizes, the corresponding passive interfaces are generated automatically via inverse EM synthesis. This allows it to close the loop from performance targets to a final layout in an end-to-end automated flow. Because the RL agent evaluates these interactions through EM-aware synthesis of the full circuit, it can make architecture-level decisions that are directly compatible with silicon implementation.

The results in Figure 14 illustrate this capability [51]. A 100 to 120 GHz PA synthesized through this pipeline

achieved 12.6-dBm output power and 12.9-dB gain using a two-stage architecture with unconventional, QR-code-like passive structures; the agent learned that power combining would reduce efficiency at these frequencies and omitted it. In contrast, a broadband 34- to 70-GHz design produced 18.5- to 21.2-dBm output power, with the agent determining that combining and dual driver stages were essential to sustain high efficiency across the band. It can be noted that final architecture, circuit topology, parameters, biasing, and sizing, including

the EM inverse synthesis, all came from the framework of combining RL and inverse design to produce an “almost” fabrication-ready RFIC. The actual connections to pads and bypass capacitors still needed to be incorporated, but that process is easy to automate. Figure 15 shows again two LNA structures, once classical looking and once more complex looking, both arising from algorithms, capturing designer intents [65].

While a first-pass design can be generated in minutes, adding practical constraints, such as bias routing,

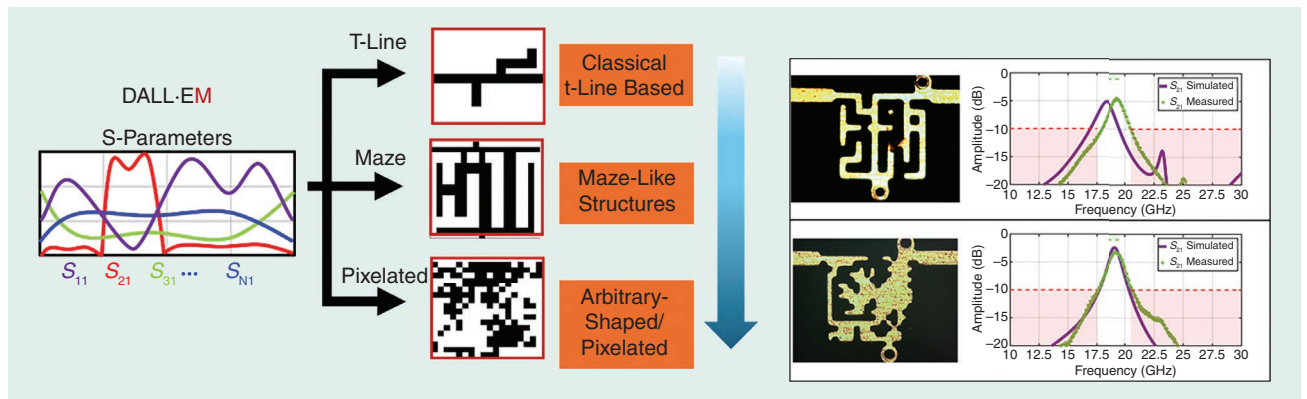


FIGURE 12: Inverse synthesis of EM structures with generative AI. Here, controlled diffusion is employed to guide the evolution of the structure toward meeting the performance objectives, where the nature of the final structure can be controlled by the designer. This can range from more familiar-looking t-lines to maze to fully pixelated and arbitrary shaped [50].

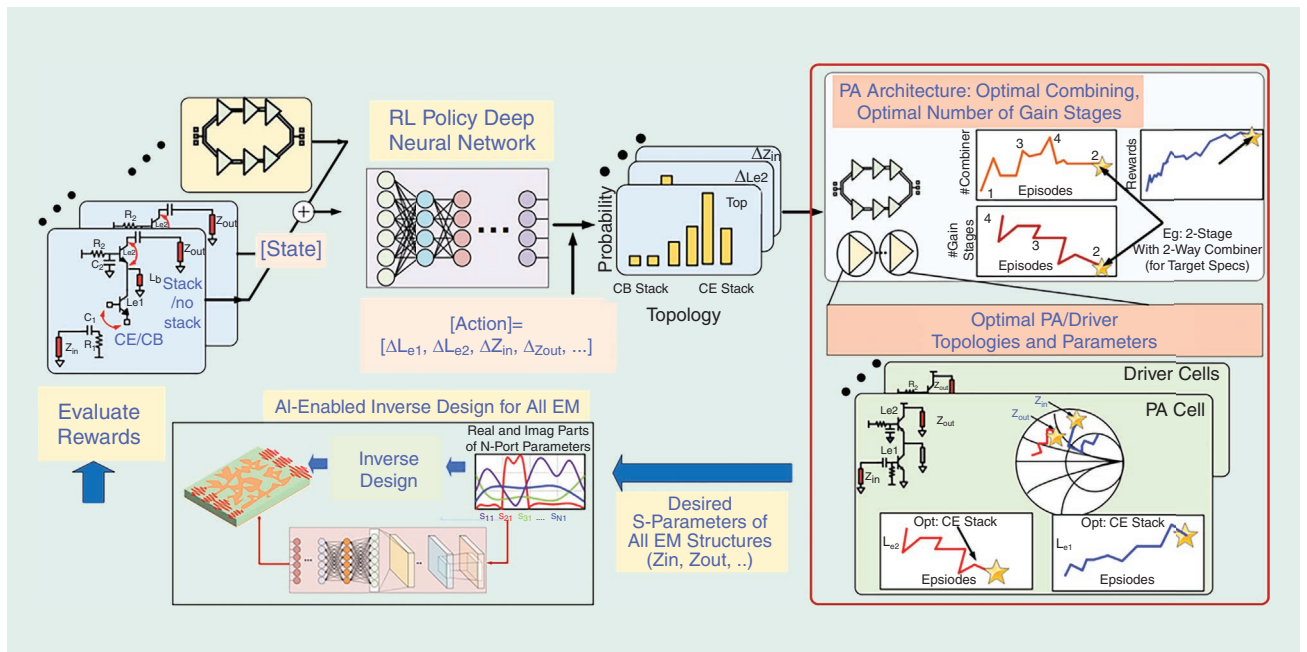


FIGURE 13: A step toward end-to-end RFIC design that combines RL and inverse design to jointly navigate architecture, circuit topologies, EM structures, and parameters. Trained on fully laid-out active devices, this approach provides a pathway to fabrication-ready layouts, enabling a fully automated specification-to-GDS flow [51].

floor-planning, and power distribution, naturally increases design time. Even then, the approach remains orders-of-magnitude faster than traditional human-driven RFIC design flows. Although demonstrated here for mmWave/subterahertz PAs, the same framing can extend to other blocks, such as voltage-controlled oscillators, LNAs, and mixers, and ultimately to higher-level RF front ends and transceivers.

So, What Did We Accomplish in the End? What Else Can You Do if Designs Can Be Done so Rapidly?

Figure 14 presents the first results for full end-to-end automated RFIC

synthesis that navigates architecture, circuit topology, and EM and the parameter space simultaneously to produce near fabrication-ready GDS from specifications.

This is not trivial at all, since it shows at least the ability (albeit in a limited setting) for algorithms to navigate the design space an experienced designer does in years of experience and weeks of design time. It also shows that EM synthesis can be algorithmized through inverse design. Antenna can be as well, with the caveat that the context needs to be well defined and understood (such as PCB/chip stacks, etc.), which is typically the case [52]. Second, algorithms

can be trained to understand the tradeoff space between topology, circuit configuration, parameters, and EM. Both of these can be combined to produce RFICs. While we demonstrate this at mmWave with PAs and LNAs with distributed EM structures, one can do this at lower RF frequencies and with other circuits.

When designs can be generated in minutes, the nature of design-space exploration changes fundamentally. As illustrated in Figure 15, rapid synthesis enables the approximate Pareto front of LNAs to be mapped quickly, giving designers a quantitative view of what is truly achievable in a given technology stack. Rather than relying

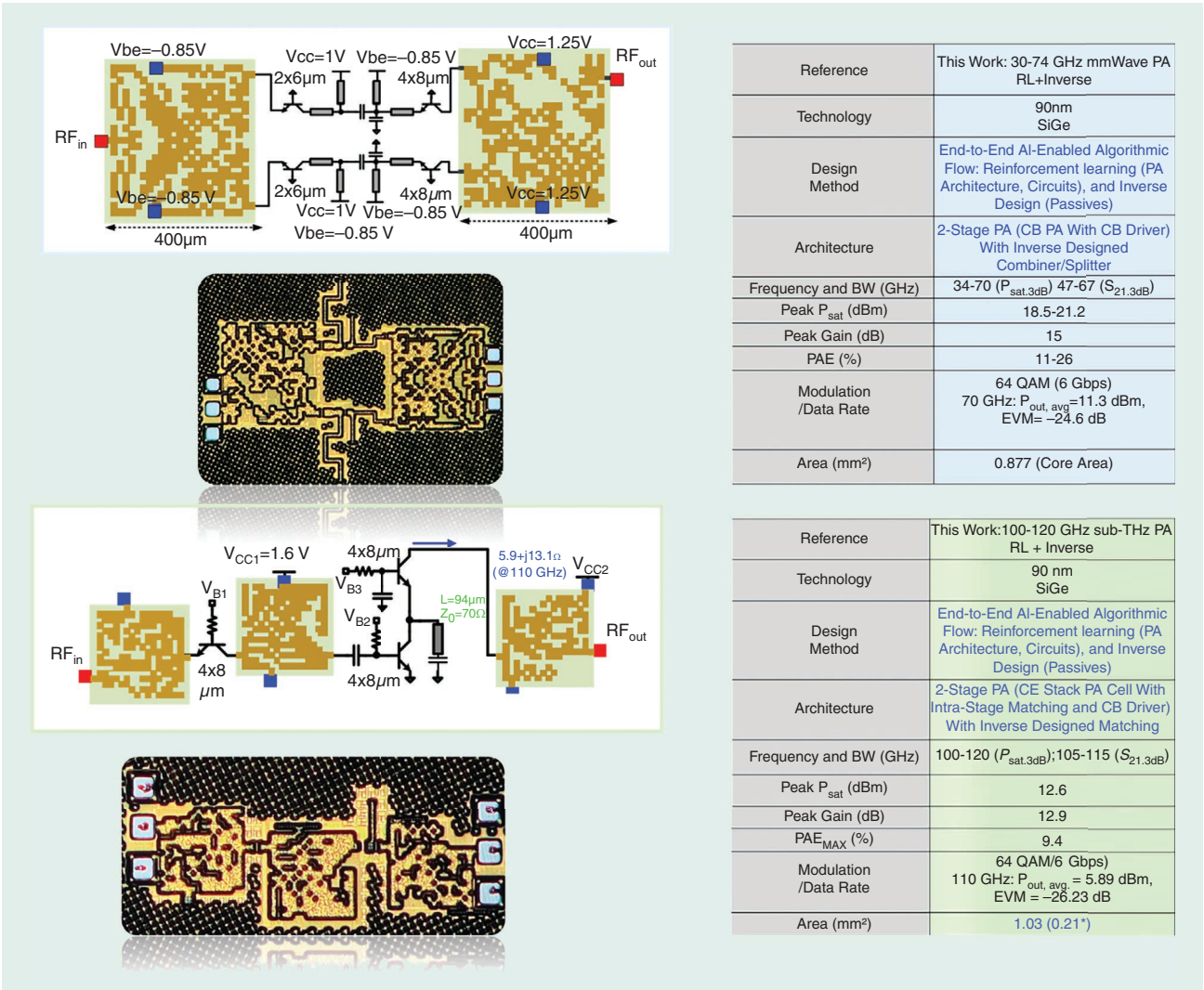


FIGURE 14: Example of a mmWave/sub-THz PA synthesized by combining RL and inverse design, enabling joint exploration of architecture, topology, EM structures, and parameters in a spec-to-GDS flow. The figure shows two examples where the algorithm determines the architecture (e.g., whether power combining is required and the number of driver stages), along with the circuit and driver topology, all design parameters, and the interface EM structures in an end-to-end manner [51].

on idealized metrics, such as $NF_{\min}$, load-pull contours, or $G_{\max}$, this approach reveals what is actually synthesizable once parasitics, topology, architecture, and EM effects are all accounted for. While inverse design cannot guarantee globally optimal matching efficiency, prior work suggests that solutions obtained in this highly nonconvex space are often close to the global optimum, albeit without providing insight into how such optimal structures can be systematically designed [66].

Challenges Ahead of Us to Fully Explore Impact of AI in RFIC

The results are extremely promising, but the answer to how much AI can influence RFIC is still unfolding. There are some significant challenges on the way, including the following:

- **Data scarcity and closed PDKs:** AI models are limited by the lack of large, shareable RFIC datasets. Proprietary PDKs under nondisclosure agreements and scarce experimental data force reliance on synthetic simulations, slowing progress in the absence of an “ImageNet for RFICs.”
- **Lack of foundational circuit models:** Most AI tools today are customized to specific nodes and processes. A true foundational

model capable of reasoning across technologies and abstraction levels has yet to emerge.

- **Optimization, benchmarking, and verification:** We lack standardized benchmarking frameworks for IC design algorithms, akin to those used for frontier LLMs, which prevents consistent and regular evaluation across approaches. The work in [53] is a good starting point, but we need more comprehensive analysis.
- **Generalizability and the “last mile”:** Designs that are merely “close” are insufficient. AI tools must generalize across problems while addressing the extremely demanding ‘last-mile’ challenges for state-of-the-art, tapeout-ready RFICs.

These challenges are real and nontrivial, yet they are not beyond reach. Addressing them will require true multidisciplinary collaboration, where circuit designers and AI scientists work side by side, shaping tools and methods together.

What Is the Role of a Circuit Designer in the World of AI?

Let us, for the sake of argument, imagine that these challenges are overcome collectively, and design methodologies leveraging AI reach to a level where it can act as a col-

laborator in its true sense (and not merely a brute-force optimizer). Where does human creativity, then, find its greatest impact? What is the role for the human circuit designer?

T.S. Elliot, in his 1934 play *The Rock*, famously writes “Where is the wisdom we have lost in knowledge? Where is the knowledge we have lost in information?” as a critique of modern civilization’s tendency to prioritize information gathering over deeper understanding and wisdom. Maybe this is the shift we should welcome. When trivial/manual optimization fades into the background, and the designer’s role shifts toward higher-level creativity and architectural insight, inviting us to rethink how RFICs may be conceived as these capabilities mature. Perhaps we can rethink specifications fully exploring new capabilities of the nonintuitive design space enabled by AI. Perhaps AI could help suggest promising circuit topologies sometimes guided by the designer, sometimes discovered on its own. It might help us see the real Pareto front, giving clarity on what a given technology can and cannot do. Over time, this could grow into true multiphysics codesign, spanning EM, circuits, thermal, and mechanical effects across RF, photonics, and mixed-signal systems.

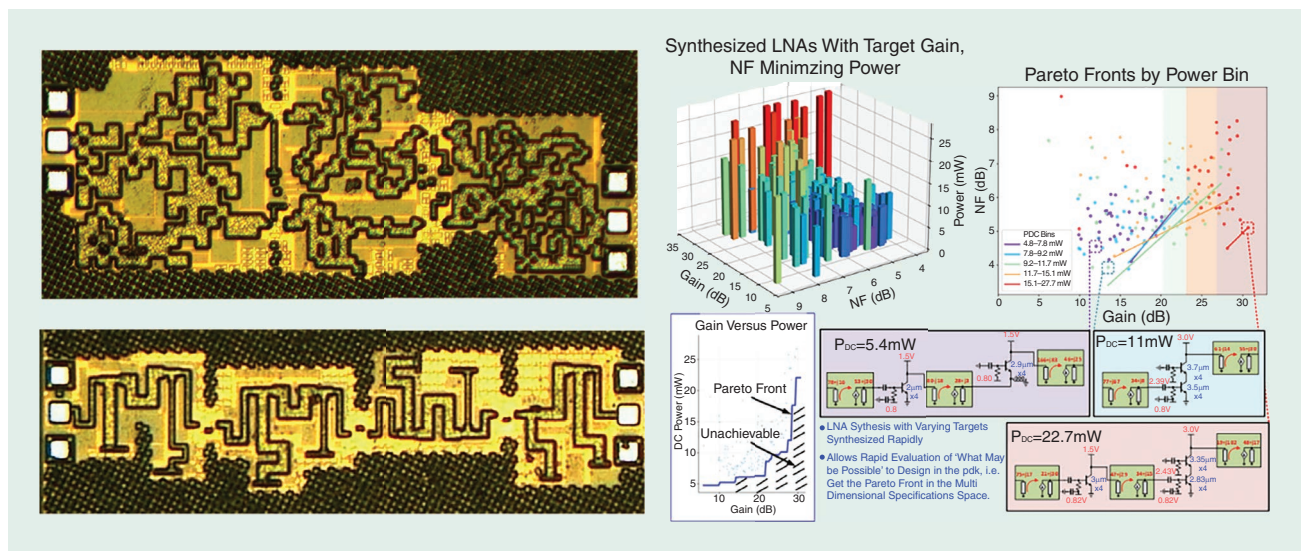


FIGURE 15: Spec-to-GDS (incorporating designer inputs): AI-enabled end-to-end design of mmWave LNAs that supports both pixelated and familiar transmission-line-like architectures, improving design interpretability. Rapid synthesis enables efficient exploration of the Pareto front to understand performance limits within a given technology node [65].

History offers a useful lens. Innovation in this space has traditionally come from deep physical intuition, reimagining how familiar building blocks can be rearranged to achieve new functionality. Think about including distributing an inductor across multiple paths in the distributed active transformer [67], codesigning transmission lines with CMOS devices for mmWave ICs [68], exploiting correlated noise paths for broadband noise-canceling LNAs [69], and using time-shifted switching in N-path circuits for spectral filtering [70]. In each case, breakthroughs arose not from new primitives, but from creative recombination of known circuit and EM elements in unconventional ways. Has AI shown the ability to do it? Not yet, but whether it can finally achieve this is not out of consideration.

A natural question arises as to whether IC designers should be cautious about embracing AI or actively engage with it. The answer is not to step away but to lean in with clarity. AI is neither a replacement for engineering judgment nor a magic solution; its current strengths lie in accelerating exploration, exposing hidden tradeoffs, and offloading repetitive optimization, while its limitations including data scarcity, verification, and generalization remain real. Designers who understand both its capabilities and its boundaries will be best positioned to shape better design flows, and ultimately, better chips.

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About the Authors

Kaushik Sengupta (kaushiks@princeton.edu) is an IEEE Fellow and currently professor in the Department of Electrical and Computer Engineering at Princeton University, Princeton, NJ 08544 USA. He received a B.Tech./M. Tech. (dual degree) in electronics and electrical communication engineering from the Indian Institute of Technology, Kharagpur, in 2007, an M.S. degree in electrical engineering from the California Institute of Technology in 2008, and a Ph.D. degree in electrical engineering from the California Institute of Technology in 2012. His research interests include novel chip-scale architectures for intelligent sensing and communication for a wide range of emerging applications. He received the Defense Advanced Research Projects Agency Young Faculty Award in 2018, the Bell Labs Prize in 2017, the Young Investigator Program Award from the Office of Naval Research in 2017, the Prime Minister Gold Medal Award from the Indian Institute of Technology (IIT) Kharagpur in 2007, the Charles Wilts Prize at Caltech for the best Electrical Engineering Ph.D. thesis in 2013, and the inaugural Young Alumni Achievement Award from IIT Kharagpur in 2018. He served as a Distinguished Lecturer for the IEEE Solid-State Circuits Society from 2019 to 2020 and for the IEEE Microwave Theory and Technology Society from 2021 to 2023. He is a recipient of the 2021 IEEE Microwave Theory and Technology Outstanding Young Engineer Award and the 2022 IEEE Solid-state Circuits New Frontier Award. He received the IEEE Microwave Prize in 2015, several best paper awards, including IEEE IMS (2020, 2021, 2022, 2025), RFIC (2012), and the Best Paper of the year award from IEEE Journal of Solid-State Circuits in 2023 for the first deep-learning enabled RFIC design. He serves as an advisor for a startup Guru Inc., based in California, focusing on long-distance wireless power transfer.

Jonathan Zhou (jz0110@princeton.edu) is a Ph.D. candidate in electrical and computer engineering at Princeton University, Princeton, NJ

08544 USA, advised by Prof. Kaushik Sengupta. His research focuses on artificial intelligence-enabled design automation for millimeter wave and subterahertz integrated circuits, leveraging reinforcement learning and inverse electromagnetic methods for end-to-end RFIC synthesis. He is a recipient of the Qualcomm Innovation Fellowship and the Princeton School of Engineering and Applied Sciences Graduate Student Award for Excellence. He received his B.S. degree in electrical engineering from the University of Washington, Seattle, in 2022.

Juho Park (jp3327@princeton.edu) is a postdoctoral research associate at Princeton University, Princeton, NJ 08544 USA. He received his B.S. degree in electrical and computer engineering from Seoul National University, Seoul, South Korea, in 2017, and his Ph.D. degree in electrical engineering from KAIST, Daejeon, South Korea, in 2024. He is a recipient of the KAIST Best Dissertation Award (2024), and was selected for Sejong Science Fellowship (2023), and Nurturing Next-Generation Researchers Fellowship (2024) from the National Research Foundation of Korea. Before joining Princeton University, he worked as a process integration engineer at Samsung Electronics, Hwaseong, South Korea. His current research focuses on inverse design methodologies for RFICs and photonics. He is a Member of IEEE.

Emir Ali Karahan (karahan@princeton.edu) received his B.S. degree from Bilkent University, Ankara, Turkey, in 2019, and his Ph.D. degree from Princeton University, NJ, USA, in 2025. His research focused on millimeter wave/RFIC design and artificial intelligence-assisted algorithmic design methods for RF/millimeter wave circuits. He was a corecipient of the Advanced Practice Paper Award at IMS 2022 and received the Princeton Yan Huo '94 Fellowship (2023), the Princeton School of Engineering and Applied Sciences Award for Excellence (2024), the Solid-States Circuits Society Predoctoral Award (2024–2025), and the JSSC Best Paper Award (2023).

Since July 2025, he has been with Marvell Technology, Santa Clara, CA 95054 USA as an analog design engineer. He is a Member of IEEE.

Jiajun Tang (jt1488@princeton.edu) received his B.S. degree from School of Electronics Engineering and Computer Science, Peking University, in July 2024. He is currently a Ph.D. student in the Department of Electrical and Computer Engineering, Princeton University, Princeton, NJ 08544 USA. His research interest spans broadly in design methodology in RF and analog integrated circuits and artificial intelligence-aided circuit design.

Sherif Ghozzy (sghozzy@princeton.edu) received his B.S. and M.S. degrees in electrical engineering from Ain Shams University, Egypt. Prior to his doctoral studies, he gained more than four years of industry experience in power electronics and analog/mixed-signal circuit design. He is currently pursuing his Ph.D. in electrical and computer engineering at Princeton University, Princeton, NJ 08544 USA, where he began in 2020. His research spans RF and analog circuit design, microwave engineering, machine learning, and optical/photonic systems, with a focus on developing energy-efficient terahertz receivers and sensors, as well as artificial intelligence-assisted methodologies for RFIC design. He is a corecipient of the 2025 Qualcomm Innovation Fellowship.

Muhamed Allam (ma0810@princeton.edu) received his B.S. and M.S. degrees in electrical engineering from Ain Shams University, Egypt, where he specialized in high-speed serial link receivers. He has more than four years of industry experience in analog and mixed-signal IC design. He is currently a Ph.D. student in electrical and computer engineering at Princeton University, Princeton, NJ 08544 USA. His research focuses on RFIC and millimeter wave circuit and system design, including reconfigurable intelligent surfaces and artificial intelligence-assisted design methodologies.

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